LIBRARY MODULE (RESOURCE MANAGEMENT ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Librarians • Teachers • IT Team • Students • Parents
Related Modules: Student Management • Academics • E-Learning • Finance • Notifications • Parent Portal

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1. MODULE OVERVIEW

1.1 Purpose

The Library Module manages all physical and digital learning resources within the school ecosystem. It enables:

- Cataloging and tracking of books and materials
- Borrowing, returning, and overdue management
- Member management (students and staff)
- Access to digital learning resources
- Analytics on resource usage and demand
- Integration with academic and e-learning systems

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2. CORE OBJECTIVES

1. Maintain a centralized catalog of all library resources
2. Track borrowing, returns, and overdue items
3. Automate fines, reminders, and reservations
4. Provide insights into resource usage and demand
5. Integrate with student and academic modules
6. Enable mobile and digital resource access
7. Ensure secure, role-based access and data protection
8. Support scalable, multi-school deployments

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Book Added → Cataloged → Available for Borrowing → Borrowed → Due Date Assigned → Returned/Overdue → Fine Calculated → Analytics Updated

3.2 Daily Operational Flow

Librarian updates catalog or inventory
↓
Student/staff searches and borrows book
↓
System records borrowing and assigns due date
↓
Automated reminders sent before due date
↓
Book returned or marked overdue
↓
Fine calculated (if applicable)
↓
Analytics and activity logs updated

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4. MODULE ARCHITECTURE

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Library Module
│
├── Catalog & Inventory Management
├── Borrowing & Returns System
├── Overdue & Fine Management
├── Reservation & Hold System
├── Member Management
├── Digital Resource Integration
├── Analytics & Reporting
├── Notifications & Alerts
├── Search & Filtering Engine
├── Mobile Access & Offline Support
├── Security & Role-Based Access
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5. CATALOG & INVENTORY MANAGEMENT

5.1 Features

- Add and manage books and resources
- Categorize by subject, author, grade level
- Track total copies and availability
- Identify missing or damaged items

5.2 Data Model

Book

- book_id
- title
- author
- subject
- isbn
- total_copies
- available_copies
- status

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6. BORROWING & RETURNS SYSTEM

6.1 Features

- Issue books to students or staff
- Track borrowing history
- Assign due dates automatically
- Support multiple borrowings per member

6.2 Data Model

Borrowing

- borrowing_id
- book_id
- member_id
- issue_date
- due_date
- return_date
- status

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7. OVERDUE & FINE MANAGEMENT

7.1 Features

- Automatically detect overdue items
- Calculate fines based on delay duration
- Send reminders and alerts
- Track fine payments (integration with Finance Module)

7.2 Rules

IF return_date > due_date
→ Mark as overdue
→ Calculate fine
→ Trigger notification

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8. RESERVATION & HOLD SYSTEM

8.1 Features

- Allow users to reserve unavailable books
- Maintain queue for high-demand resources
- Notify users when book becomes available

8.2 Data Model

Reservation

- reservation_id
- book_id
- member_id
- queue_position
- status

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9. MEMBER MANAGEMENT

9.1 Features

- Track students and staff as library members
- Link to Student and Staff modules
- Monitor borrowing limits and activity

9.2 Data Model

Member

- member_id
- type (student/staff)
- linked_user_id
- active_status

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10. DIGITAL RESOURCE INTEGRATION

10.1 Features

- Access external educational platforms (e.g., OER, e-books)
- Link resources to subjects and curriculum
- Provide downloadable or online materials

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11. SEARCH & FILTERING ENGINE

11.1 Features

- Search by title, author, subject, ISBN
- Advanced filters (year, availability, category)
- Fast query performance for large catalogs

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12. ANALYTICS & REPORTING

12.1 Features

- Borrowing trends over time
- Most popular books and subjects
- Member activity reports
- Overdue and fine statistics

12.2 Dashboards

- Library usage overview
- Resource demand insights

- Active vs inactive members

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13. NOTIFICATIONS & ALERTS

13.1 Features

- Due date reminders (SMS/email/push)
- Overdue alerts
- Reservation availability notifications

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14. MOBILE-FIRST & OFFLINE DESIGN

14.1 Requirements

- Responsive mobile interface
- Search and borrowing via smartphones
- Offline catalog access with sync

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15. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full control, analytics, configuration
Librarian	Manage catalog, borrowing, fines
Teacher	Borrow/view resources
Student	Search, borrow, reserve books
Parent	View child’s borrowing activity

15.2 Security Features

- Secure authentication and authorization
- Data encryption
- Audit logs for all transactions

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16. SCALABILITY

- Support thousands of books and users
- Multi-campus and multi-tenant architecture
- Optimized search and reporting performance

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17. IMPLEMENTATION GAPS (CRITICAL FIXES)

17.1 Current Issues Identified

- No advanced search and filtering
- Missing fine/fee automation
- No reservation/hold system
- Limited analytics and reporting
- Lack of automated notifications
- Weak role-based access visibility

17.2 Required Fixes

- Implement advanced search and filters
- Add automated fine calculation and payment tracking
- Introduce reservation queue system
- Build analytics dashboards
- Enable automated notifications (SMS/email)
- Strengthen role-based access and audit logs

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18. FUTURE ENHANCEMENTS

- Barcode/QR code scanning for book tracking
- RFID-based inventory management
- AI-powered book recommendations
- Integration with national digital libraries
- Gamification (reading points, badges)
- Voice search and smart assistants